Detection Without Identification: ShotSpotter and Gun Homicide in Chicago, 2009–2023

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Abstract

Between 2017 and 2023, Chicago paid millions of dollars to deploy ShotSpotter—an acoustic gunshot-detection system—across 51 of its 77 community areas, and Mayor Brandon Johnson cancelled the contract in 2023 citing limited evidence of impact. That ShotSpotter does not reduce gun violence is by now well established—in Chicago at the police-district level (Connealy et al., 2024) and nationally (Doucette et al., 2021); the question we take up is not whether that null replicates but whether the public, community-area-level data that anchor most policy debate can *identify* a causal effect at all. Using a community-area-by-year panel of gun homicides spanning 2009–2023, we estimate five classes of model: (i) a two-way fixed-effects OLS difference-in-differences (DiD); (ii) Poisson and Negative Binomial regressions appropriate for count outcomes; (iii) cohort-specific and Callaway–Sant'Anna (2021) event studies that respect the staggered 2017/2018 rollout; (iv) a synthetic control with permutation inference; and (v) pre-period placebo tests. The OLS DiD coefficient is positive and significant ($\beta = +2.63$ gun homicides per area-year, $p < 0.001$) and survives unit and year fixed effects—but it is a count-data misspecification artifact (Santos Silva & Tenreyro, 2006): estimated as Poisson or Negative Binomial with the same fixed effects, the effect collapses to a statistically null incidence-rate ratio of ≈ 1.07 ($p \approx 0.5$), with confidence intervals spanning anywhere from a 13% reduction to a 33% increase; a heterogeneity-robust Poisson extended two-way fixed-effects estimator (Wooldridge, 2023) returns the same null, so the result is not a fixed-effects weighting artifact. The design also fails a joint parallel-trends Wald test ($F = 19.9$, $p = 0.006$), produces similarly "significant" coefficients when treatment is faked in pre-rollout years, and admits no feasible synthetic control because the 26 never-treated areas are systematically lower-violence than any treated neighborhood. A placement model makes the selection explicit: ShotSpotter siting is predicted by structural disadvantage and pre-period violence with near-perfect separation (propensity AUC 0.94), and the six highest-violence study neighborhoods have no comparably situated untreated counterpart. These data therefore *cannot identify* a causal effect of ShotSpotter on gun homicides in either direction. The installation estimates nonetheless rule out a large *deterrence* effect, but cannot exclude the survival-channel mortality benefit that a boundary regression discontinuity attributes to faster emergency response—a magnitude that lies inside our confidence interval and that this design is not built to detect. The same non-identification recurs on an independent event—Chicago's September 2024 shutoff, where gun homicides were already falling city-wide in coverage and comparison areas alike, so the widely cited post-removal decline reflects a city-wide trend, not a removal effect. The contribution is thus a worked identification-limits case: a coefficient can survive fixed effects and robustness restrictions and still fail every condition a causal reading requires.

1 Introduction

Gun violence in the United States is a persistent public-health crisis that disproportionately affects young people of color and is heavily concentrated in large metropolitan areas (CDC, 2020; Kegler, Dahlberg, & Mercy, 2018). One barrier to reducing firearm violence is the chronic underreporting of gunfire: research finds that only about 20% of shootings are reported to law enforcement (Irvin-Erickson et al., 2016; Carr & Doleac, 2016). In response, cities have turned to gunshot-detection technology—most notably ShotSpotter—to close that measurement gap and improve real-time awareness of gunfire (Choi, Librett, & Collins, 2014). This paper uses Chicago's deployment of that technology not to re-evaluate it, but to ask what the public data most policy debate relies on can—and cannot—identify.

Chicago piloted ShotSpotter in 2012 in two small zones, expanded to full district coverage in 2016, and signed a three-year, $3 million contract to widen coverage in 2018 (Office of Inspector General, 2021). The system has been controversial: a 2021 Office of Inspector General (OIG) audit found that alerts rarely produced evidence used in investigations; advocacy groups argued the technology concentrated police presence in already over-policed Black and Brown communities; and in 2023 Mayor Brandon Johnson terminated the contract, citing reliability and community-impact concerns.

The empirical question that motivated ShotSpotter is straightforward—*did it reduce gun violence?*—and for Chicago it is largely answered. A matched quasi-experiment of the technology's staggered rollout across Chicago police districts finds no reduction in fatal or non-fatal shootings, gun crimes, or shots-fired calls, only a rise in recovered evidence (Connealy et al., 2024); a national panel of large metropolitan counties reaches the same null for firearm homicides and arrests (Doucette et al., 2021); and a district-level study finds that, if anything, the technology slowed dispatch and lowered arrest probability (Topper & Ferrazares, 2024). Replicating that null is not this paper's contribution. What no prior evaluation has done is run the design at the community-area level—the 77-unit geography in which Chicago's open data are published and its policy debate is conducted, finer than the police districts of the existing Chicago evaluations (Connealy et al., 2024; Topper & Ferrazares, 2024).

Our question is instead a methodological one: can the public, community-area-level data that dominate policy debate *identify* such an effect at all, and what happens when a conventional evaluation is run on them without asking? The answer, defensibly stated, is that **no design we can run produces a credible causal estimate**—and the instability is diagnostic, not incidental. It is the signature of a research design with no valid control group and a pre-period dominated by a single anomalous year. Three findings organize the paper. First, a standard two-way fixed-effects DiD manufactures a large, significant, robustness-surviving positive coefficient that collapses to a null the moment the count outcome is modeled correctly (Santos Silva & Tenreyro, 2006; Wilson, 2022)—a worked instance, on a live $3 million procurement decision, of a coefficient that passes every routine check yet fails a causal reading (LaLonde, 1986). Second, we estimate the selection mechanism the design founders on: ShotSpotter placement is predicted by structural disadvantage and prior violence so tightly that the highest-violence treated neighborhoods have no comparably situated untreated counterpart, which is what makes a synthetic control literally non-constructible rather than merely imperfect. Third, we state the substantive envelope honestly: the data rule out a large *deterrence* effect but cannot exclude a survival-channel mortality benefit of the kind a boundary regression discontinuity attributes to faster response—a magnitude that falls inside our own confidence interval. The same two failures resurface when we treat Chicago's September 2024 removal of the system as an independent second experiment—a city-wide decline mistaken for a treatment effect, and an additive coefficient a count model erases—confirming that the non-identification is structural rather than an accident of how the installation happens to be timed. We close with the conclusion the data robustly support—that ShotSpotter functioned as a detection instrument, not as a demonstrated violence-reduction intervention—a reading consistent with the 2023 decision to end the contract.

2 Background: ShotSpotter in Chicago

ShotSpotter is a real-time acoustic-detection service. Microphones triangulate the sound of gunfire, an algorithm filters non-gunshot sounds, and a human acoustic expert reviews the alert before it is dispatched to police (Choi, Librett, & Collins, 2014; Gatens & Reichert, 2019). The technology is sold as a response-time tool: faster, more accurate detection should mean faster police response and faster trauma care, and therefore—in principle—fewer deaths. While some scholars argue that policing and enforcement can reduce gun violence (Braga & Cook, 2023), others caution that aggressive enforcement carries serious social costs for communities already disproportionately affected by violence (Harcourt, 2019; Raphael & Stoll, 2013).

The evaluation record on gunshot detection is, by now, fairly settled, and it splits along one seam: the technology reliably improves *process* outcomes—faster detection, more ballistic evidence, fewer shots-fired calls—but not the *victim* outcomes that motivate its purchase. A microsynthetic-control evaluation in Kansas City draws the seam sharply (Piza et al., 2024): coverage raised ballistic-evidence collection and cut shots-fired calls, yet left fatal shootings, non-fatal shootings, and gun assaults unchanged. Neighborhood evaluations in St. Louis echo it—roughly a 25% drop in citizen shots-fired calls with no change in violent crime (Mares & Blackburn, 2021)—while results elsewhere range from crime reductions (Mares, 2023) to increases coinciding with deployment (Vovak et al., 2021). The same null holds at coarser units: staggered evaluations across Chicago police districts (Connealy et al., 2024) and metropolitan counties nationally (Doucette et al., 2021) find no reduction in shootings, homicides, or case clearance, a Chicago district study finds detection even slowed dispatch and lowered arrest probability (Topper & Ferrazares, 2024), and a 2026 meta-analysis of 44 estimates across eight studies pools to a null relative incidence-rate ratio of 1.02 (95% CI $[0.90, 1.16]$; Huff, Dunlap, & Pearson, 2026). The one channel with a credible claim to benefit is medical: a boundary regression discontinuity attributes a lower shooting-fatality rate inside coverage to faster emergency response (University of Chicago Crime Lab, 2024, not peer-reviewed). Against that backdrop, the question this paper takes up is not whether detection deters homicides—the weight of the evidence says it does not—but whether the community-area data on which most local policy argument relies can *identify* such an effect at all, and how a routine evaluation run on those data can mislead.

Chicago's deployment was concentrated in the city's most disinvested neighborhoods. Each of the six neighborhoods studied here—Austin, Humboldt Park, North Lawndale, Englewood, West Englewood, and South Shore—has a poverty rate roughly twice the Chicago average (28.6%–46.6% versus 19.7% city-wide), unemployment 1.5–3 times the city rate, and a hardship index between 55 and 94 against a city-wide median below 50 (Census 2008–2012). The 51 ShotSpotter community areas accounted for an average of 381 gun homicides per year in 2009–2016, against 42 per year across the 26 community areas without ShotSpotter—roughly a 9-to-1 ratio in aggregate, and about 4.6 times the homicides *per area*. That ratio is the central methodological challenge of the paper: **ShotSpotter was deployed *because* of concentrated violence and disadvantage, not assigned at random**—a selection rule we estimate directly in Section 5.6. Figure 1 shows the resulting geography of treatment and control.

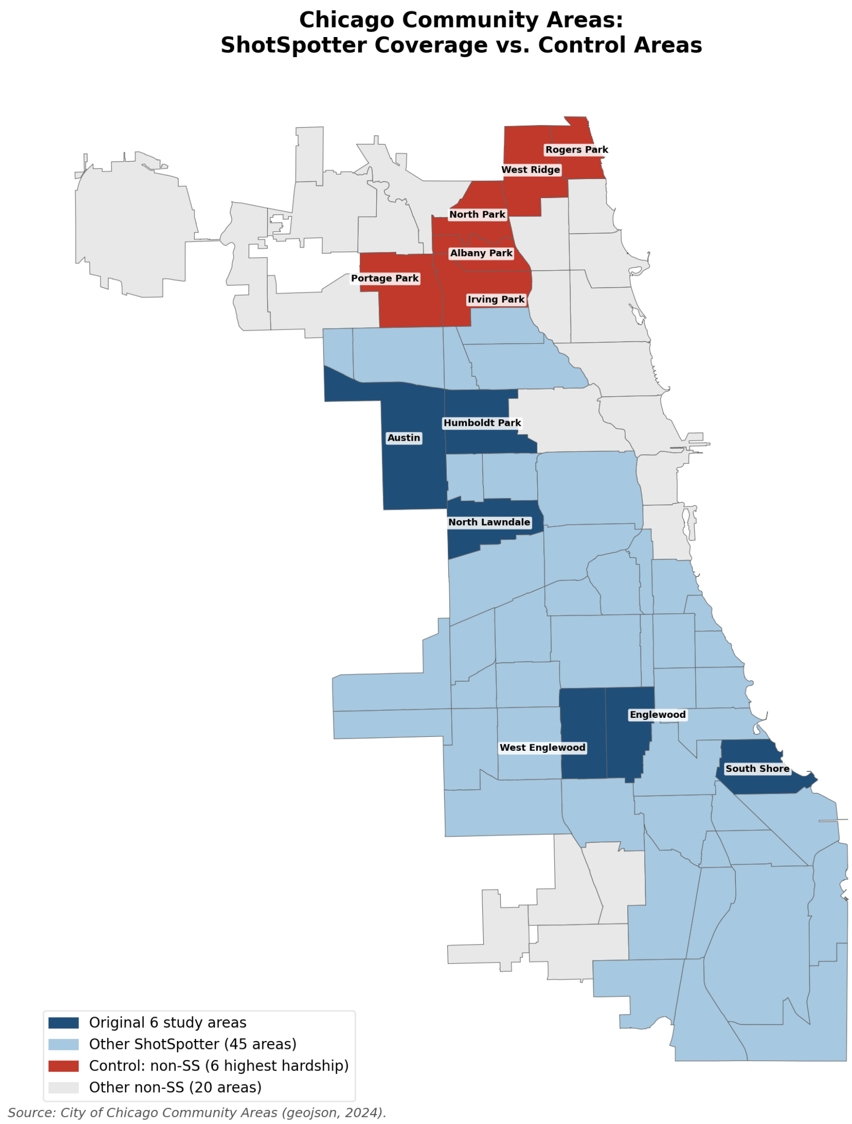

Figure 1: ShotSpotter coverage (treatment, blue) blankets the high-violence South and West sides; the 26 never-treated community areas form the control group, with the six highest-hardship ones highlighted in red. Treatment was assigned because of pre-existing violence, which is the selection problem every design in this paper must confront.

3 Data

The analysis combines four City of Chicago open-data products, all freely available from the Chicago Data Portal:

- **Violence Reduction—Victims of Homicides and Non-Fatal Shootings** (61,450 incidents, 1991–2025), filtered to gun homicides ($n \approx 7{,}600$ in 2009–2023) and non-fatal gunshot injuries ($n \approx 30{,}000$).

- **Violence Reduction—ShotSpotter Alerts (Historical)** (222,108 alerts, 2017-01-13 to 2024-09-22, when the city ended ShotSpotter; the 191,824 alerts through 2023 fall in our analysis window).
- **Census Data—Selected Socioeconomic Indicators, 2008–2012** (77 community areas plus the city total).
- **ACS 5-Year Data by Community Area** (racial composition) and **Public Health Statistics—Life Expectancy by Community Area** (2010), used in the placement model.
- **Crimes—2001 to Present** (`ijzp-q8t2`), aggregated to community-area-by-year weapons-violation incidents and arrests for the procedural-channel analysis.
- **Community Areas GeoJSON** (77 polygons).

The ShotSpotter dataset reveals the staggered rollout structure that the analysis exploits: 21 community areas appear in the data beginning in early 2017 and 30 between mid-2017 and mid-2018, leaving 26 never-treated areas. Treatment status is inferred from alert presence, so a community area counts as treated when district-level sensor coverage reached any part of it. The original project pipeline collapses both cohorts to a single 2017 cutoff—the standard convention, but one that biases pooled estimates when treatment timing varies (Goodman-Bacon, 2021), motivating the cohort-aware estimators below. Homicide and alert data are aggregated to the community-area-by-year (and, for descriptive figures, community-area-by-month) level.

Figure 2 shows the raw before/during homicide comparison for the six study neighborhoods. Counts rose in most of them—but, as the rest of the paper makes clear, a raw before/during comparison is exactly the kind of contrast that selection and the city-wide trend render uninterpretable as a causal effect.

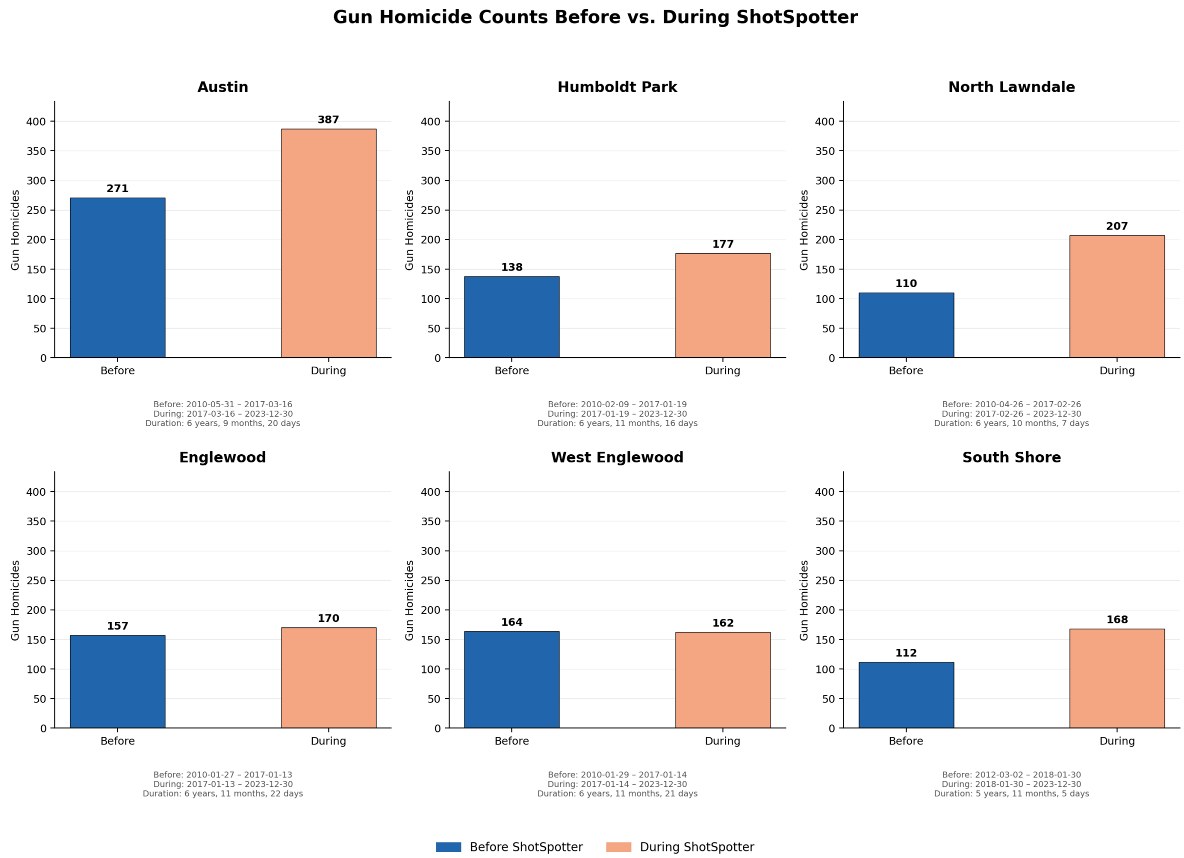

Figure 2: Total gun homicides before versus during ShotSpotter in each study neighborhood. Raw counts rose in most neighborhoods, but this naive before/during contrast confounds the intervention with the city-wide violence trend and with selection into treatment.

4 Empirical Strategy

4.1 The identification problem

In a randomized experiment we would compare treated neighborhoods to a control group drawn from the same distribution of pre-treatment violence. Chicago's data do not allow that. Because the 51 ShotSpotter neighborhoods had roughly nine times as many gun homicides in aggregate as the 26 non-ShotSpotter neighborhoods—about 4.6 times as many per area—*before* deployment, any difference-in-differences contrast must lean on the parallel-trends assumption: that, absent ShotSpotter, the two groups would have moved in parallel. Much of the paper is a sustained test of whether that assumption can survive contact with the data.

4.2 Specifications

We estimate five classes of model.

1. **Two-way fixed-effects DiD.** $\text{gun\_hom}_{it} = \beta\,(\text{treat}_i \times \text{post}_t) + \alpha_i + \gamma_t + \varepsilon_{it}$, with cluster-robust standard errors at the community-area level, estimated on the full panel and with three robustness restrictions (excluding 2016, excluding the COVID years 2020–2021, and excluding both).

2. **Count-data regression.** Poisson and Negative Binomial DiD with the same two-way fixed effects, appropriate for integer homicide counts. The Negative Binomial dispersion is *estimated* (Cameron–Trivedi NB2 auxiliary regression; Cameron & Trivedi, 2013) rather than fixed at one. As a heterogeneity-robust count analog of the Callaway–Sant'Anna estimator—and a check that the count-model null is not a fixed-effects weighting artifact—we also report a Poisson extended two-way fixed-effects model (Wooldridge, 2023), which saturates the treatment in cohort and calendar time and aggregates cohort-by-year effects from clean never-treated comparisons.

3. **Event studies.** A transparent, hand-rolled cohort-specific $\text{ATT}(g,k)$ estimator (2017 and 2018 cohorts versus the 26 never-treated areas, sample-weighted, bootstrap standard errors) and the modern **Callaway & Sant'Anna (2021)** group-time ATT (estimated with the `differences` package, influence-function inference), which we treat as the authoritative staggered-rollout result.

4. **Synthetic control.** For each of the six study neighborhoods, weights over the 26 never-treated areas are chosen to minimize pre-treatment squared error on annual gun-homicide counts, subject to the simplex constraint (Abadie, Diamond, & Hainmueller, 2010), with in-space permutation inference.

5. **Pre-period placebo / falsification.** The DiD is re-estimated with treatment reassigned to fake years (1999, 2003, 2007, 2011) well before the actual rollout. A non-null placebo coefficient

flags spurious significance and undermines the design.

5 Results

5.1 ShotSpotter detected what it was designed to detect

Alert volume in covered neighborhoods is enormous: roughly 222,000 alerts across about seven years and 51 communities, with the heaviest-coverage areas (Austin, West Englewood, Englewood, North Lawndale) generating 8,000–15,000 alerts each. Both alert volume and homicide counts peak in the same months (May–August) and the same hours (10 PM to 1 AM); Figure 3 shows the within-day co-movement. That co-movement is expected—detected gunfire and gun homicide share the same late-night diurnal cycle—so we read the figure only as validation that the sensor network captures the gunfire it was purchased to capture, not as a finding in its own right. As a *measurement instrument*, ShotSpotter works. The question these data must answer is whether detection translated into fewer homicides—and whether they can answer it at all.

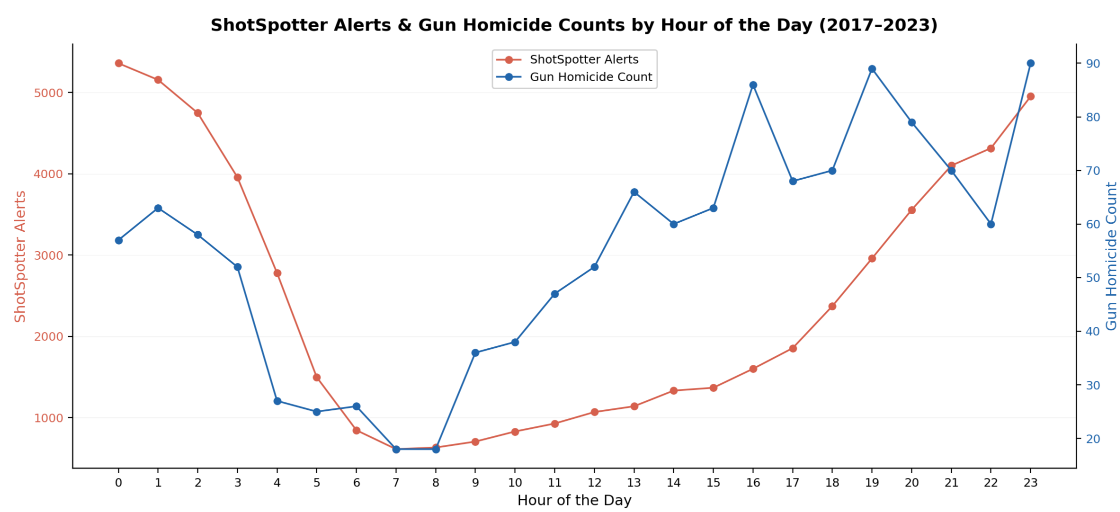

Figure 3: ShotSpotter alerts and gun homicides rise and fall together over the day, both bottoming out mid-morning and peaking late at night—an expected co-movement, shown as validation of the sensor network rather than as a finding. The system detects the gunfire it is designed to detect.

5.2 The headline OLS DiD is positive—but disappears under count-data regression

The two-way fixed-effects DiD coefficient on the full 2009–2023 panel is $\beta = +2.627$ gun homicides per community-area-year (SE $= 0.638$, $p < 0.001$, 95% CI $[+1.38, +3.88]$). It is robust across restrictions: excluding COVID years yields $+1.75$ ($p < 0.001$), excluding 2016 yields $+3.19$ ($p < 0.001$), and excluding both yields $+2.31$ ($p < 0.001$) (Table 1). Crucially, the coefficient is *robust to fixed effects*—adding unit and year fixed effects reproduces the original DiD coefficient almost exactly—so it does not come from a missing-controls problem.

Table 1: Two-way fixed-effects OLS DiD on gun homicides (cluster-robust SEs).

Specification	β (DiD)	SE	p	95% CI	N
Full panel, 2009–2023	$+2.627$	0.638	< 0.001	$[+1.38, +3.88]$	1,155
Excl. COVID (2020–21)	$+1.749$	0.502	< 0.001	$[+0.77, +2.73]$	1,001
Excl. 2016	$+3.189$	0.767	< 0.001	$[+1.69, +4.69]$	1,078
Excl. COVID + 2016	$+2.311$	0.624	< 0.001	$[+1.09, +3.54]$	924

What the headline is *not* robust to is the choice of an appropriate model for count data. Annual gun-homicide counts in a community area are small non-negative integers whose variance grows with the mean; OLS treats them as continuous and homoscedastic, weighting all observations equally and reporting an effect on the additive scale. Poisson and Negative Binomial regressions let the variance scale with the mean and estimate effects on the log scale, where they correspond to multiplicative (proportional) changes—the natural metric for a percentage-effect treatment. Estimated this way, the effect is **statistically null across every robustness specification** (Table 2): Poisson and Negative Binomial incidence-rate ratios (IRRs) hover around 1.07 with confidence intervals consistent with anywhere from a 13% reduction to a 33% increase in homicides. Because the estimated Negative Binomial dispersion is small ($\alpha \approx 0.02$)—the counts are only mildly overdispersed—the Negative Binomial IRRs land essentially on top of the Poisson estimates. They also land essentially on top of the meta-analytic pooled effect for gunshot detection (relative IRR 1.02, 95% CI $[0.90, 1.16]$; Huff, Dunlap, & Pearson, 2026), estimated with the same incidence-rate-ratio metric—evidence that the multiplicative-scale null is a property of the intervention, not an artifact of this particular panel.

Table 2: Count-data DiD with two-way fixed effects. IRR $= \exp(\beta)$; an IRR of 1 means no effect.

Specification	β	IRR	95% CI on IRR	p
OLS (TWFE), full panel	$+2.627$	—	—	< 0.001
Poisson (TWFE), full panel	$+0.072$	1.074	$[0.869, 1.328]$	0.508
Poisson (TWFE), excl. COVID	$+0.014$	1.014	$[0.816, 1.259]$	0.902
Poisson (TWFE), excl. 2016	$+0.084$	1.087	$[0.857, 1.381]$	0.491
Neg. Binomial, full panel	$+0.067$	1.069	$[0.860, 1.329]$	0.548
Neg. Binomial, excl. COVID	$+0.014$	1.014	$[0.810, 1.269]$	0.902
Neg. Binomial, excl. 2016	$+0.074$	1.077	$[0.844, 1.373]$	0.551

Figure 4 shows the contrast in one picture: the OLS effect, placed on the multiplicative scale for comparison (implied IRR ≈ 1.35, $p < 0.001$), against the genuinely estimated Poisson and Negative Binomial IRRs, both of which collapse the headline to a null near 1.07. Whatever the OLS DiD is picking up, it is being amplified by the additive scale and by the unequal variance across community areas; on the natural multiplicative scale, no effect is detectable. Figure 5 shows the same heterogeneity at the neighborhood level: individual before/after changes in the gun-homicide rate range from about -2% (West Englewood) to $+62\%$ (South Shore), and even the pooled estimate ($+23\%$, excluding COVID) carries a wide interval. For completeness, non-fatal gunshot injuries (a four-times-larger sample) yield $\beta = +7.51$ under OLS ($p = 0.009$)—and the count-data correction is even starker there: estimated as Poisson or Negative Binomial with the same fixed effects, the point estimate flips *below* one and is statistically null in every specification

(IRRs 0.84–0.92, $p \geq 0.23$; Table 3). The higher-powered outcome delivers the same verdict as the homicide series: the additive OLS coefficient is a scale artifact. The same distortion recurs, unchanged, when the identical difference-in-differences is run on weapons-violation enforcement rather than homicides—an order-of-magnitude-larger OLS coefficient that again collapses to a null incidence-rate ratio, confirming the inflation is a property of the additive specification and not of homicide counts (Appendix A).

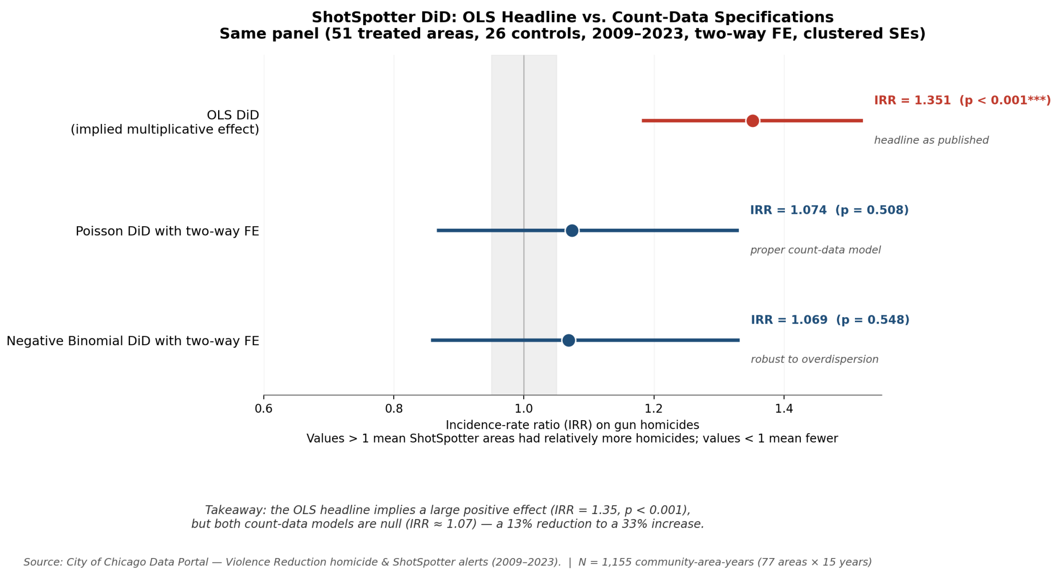

Figure 4: The whole story in one figure. The OLS effect is the additive $+2.63$ coefficient shown on the multiplicative scale for comparison (implied IRR ≈ 1.35, $p < 0.001$); the Poisson and Negative Binomial points are genuinely estimated incidence-rate ratios. Both count-data models collapse the headline to a statistically null IRR ≈ 1.07.

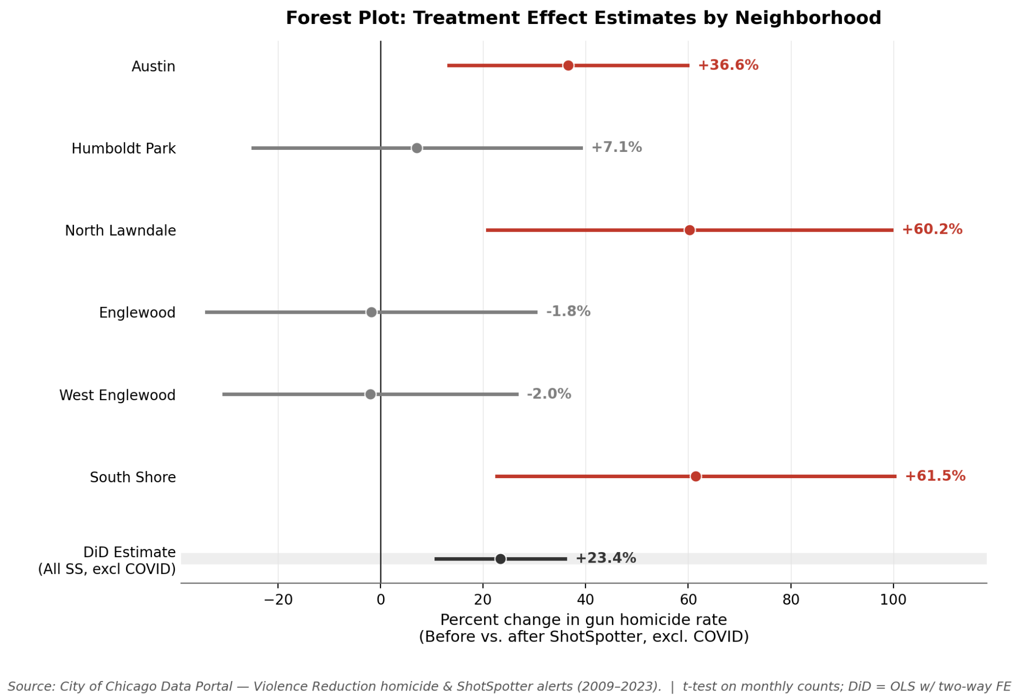

Figure 5: Neighborhood-level before/after change in the gun-homicide rate (excluding COVID), with the pooled DiD estimate at the bottom; changes whose intervals exclude zero in red, others in gray. The spread across neighborhoods—from roughly -2% to $+62\%$—shows how heavily the pooled headline leans on a few high-count neighborhoods.

Table 3: Non-fatal gunshot injuries (a four-times-larger sample) under count models, same two-way fixed effects and cluster-robust SEs. The OLS coefficient is positive and significant; every count model is null with a point estimate *below* one.

Model	Specification	β	IRR	95% CI on IRR	p
OLS (TWFE)	full panel	$+7.51$	—	—	0.009
Poisson	full panel	-0.102	0.903	$[0.691, 1.181]$	0.457
Neg. Binomial	full panel	-0.168	0.845	$[0.640, 1.115]$	0.234
Poisson	excl. COVID	-0.093	0.912	$[0.710, 1.171]$	0.469
Neg. Binomial	excl. COVID	-0.157	0.854	$[0.653, 1.118]$	0.251
Poisson	excl. 2016	-0.087	0.917	$[0.686, 1.226]$	0.558
Neg. Binomial	excl. 2016	-0.175	0.840	$[0.623, 1.131]$	0.251

5.3 The design produces “significant” placebos

If the $+2.627$ reflected a real treatment effect, then assigning *fake* treatment dates well before the actual rollout should yield null estimates. It does not (Table 4). Two of four placebos are statistically significant at the 1% level, with magnitudes (1.3–1.9 homicides per area-year) on the same order as the post-2017 estimate. They are opposite-signed, but that is precisely the point: a design that manufactures “significant” effects in years before the program existed is capturing pre-existing trend differences across neighborhoods—most likely the long-run, differentially absorbed city-wide decline from the early-1990s peak—not the intervention. That a difference-in-differences specification can return spurious significance is exactly the pathology Bertrand, Duflo, and Mullainathan (2004) documented, in which conventional standard errors so understate uncertainty that a large share of randomly dated placebo interventions test as significant. Figure 6 plots the placebo coefficients against the true post-2017 estimate.

Table 4: Pre-period placebo tests: DiD with treatment reassigned to fake pre-rollout years.

Fake treatment year	β	SE	p	95% CI
1999	-1.344	0.507	0.008	$[-2.34, -0.35]$
2003	-1.929	0.714	0.007	$[-3.33, -0.53]$
2007	-0.645	0.497	0.194	$[-1.62, +0.33]$
2011	$+0.256$	0.347	0.461	$[-0.42, +0.94]$

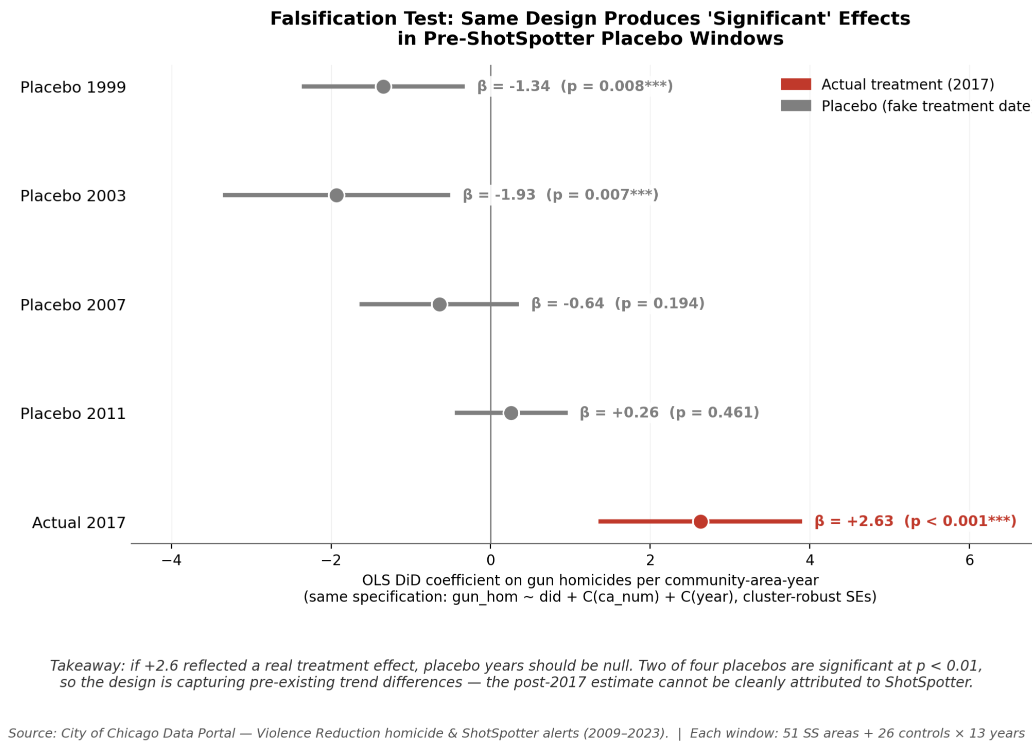

Figure 6: Falsification test. Assigning fake treatment dates before ShotSpotter existed should produce null effects; instead the 1999 and 2003 placebos come back significant at the 1% level and of comparable magnitude to the true post-2017 estimate.

5.4 The event studies confirm a failure of parallel trends

The cohort-specific event study (Appendix Table 13, aggregated across the 2017 and 2018 cohorts with never-treated controls) shows that *every* pre-treatment coefficient is large and negative—a clear failure of parallel trends. A formal joint Wald F-test on the seven pre-treatment interaction coefficients, which is invariant to the choice of base year, yields $F = 19.9$, $p = 0.006$, rejecting parallel trends at the 1% level. The pattern is mechanical: 2016 was a Chicago-wide homicide-record year, disproportionately concentrated in the neighborhoods that would soon receive ShotSpotter, and the event study uses a year adjacent to 2016 as its base, so every other year looks anomalously low (Figure 7). Once a design fails this test, the DiD contrast cannot be read as causal. This is not a technicality to be tested away: pre-trends tests are themselves underpowered, and conditioning on having "passed" one distorts the inference that follows (Roth, 2022). A design that visibly *fails* the test, as this one does, is telling us the parallel-trends assumption is not available—and the credible response is to report that the effect is not point-identified, not to lean on a single base-year normalization.

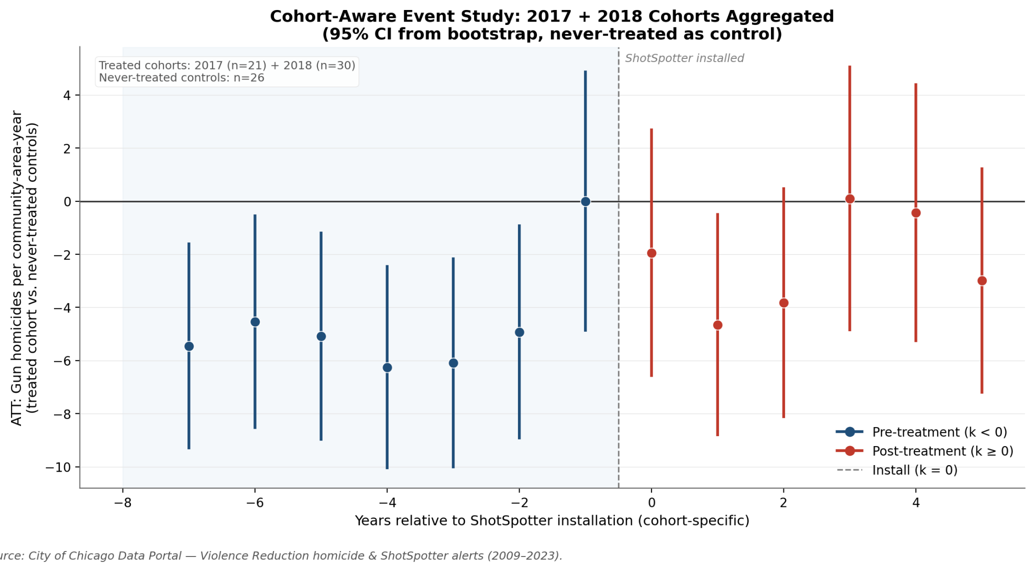

Figure 7: Cohort-aware event study with bootstrap confidence intervals. Large negative pre-treatment coefficients reveal a violation of parallel trends driven by the 2016 record-homicide year sitting next to the base period.

The hand-rolled event study is intuitive but is not the modern estimator the staggered-rollout literature prescribes. As a final check we re-estimate the effect with the **Callaway & Sant'Anna (2021)** group-time ATT (Table 5). On the full panel the overall ATT is -2.45 per community-area-year (SE 0.89)—*negative*, the opposite sign of the pooled OLS, and stable to using not-yet-treated rather than never-treated controls (-2.51). But this negative sign is not a property of the estimator. It is driven almost entirely by the 2016 record-homicide year, which sits at the $k = -1$ baseline for the 2017 cohort: the event study shows that only the $k = -1$ (2016) pre-coefficient is significant while the rest hover at zero, and **excluding 2016 and re-estimating returns** $+1.90$ **(SE** 0.72**)—the same sign as the OLS DiD** (Figure 8). The apparent OLS-versus-Callaway–Sant'Anna sign reversal is therefore a single-year sensitivity, not a sign that the estimators fundamentally disagree about a well-identified parameter. The Callaway–Sant'Anna estimator is precisely the tool the staggered-adoption literature prescribes to remove the negative weighting that can flip a two-way fixed-effects coefficient under heterogeneous timing (Goodman-Bacon, 2021; de Chaisemartin & D'Haultfœuille, 2020). That even it returns a sign contingent on a single year confirms the parameter is not identified in these data, rather than restoring it. A Goodman–Bacon decomposition of the staggered two-way fixed-effects estimate (+2.48, cohort-specific timing) locates the problem precisely: the "forbidden" comparison of the 2018 cohort against the already-treated 2017 cohort carries only 4.6% of the weight, and the weighted 2×2 components reproduce the aggregate exactly (Figure 9, Table 6). Negative-weighting contamination is therefore minor in this panel—the instability comes from the 2016 anomaly and the absence of a valid control group, not from the staggered aggregation itself.

Table 5: Callaway–Sant'Anna (2021) overall ATT, gun homicides per community-area-year.

Control group	Overall ATT	SE
Never-treated	−2.453	0.893
Not-yet-treated	−2.508	0.898
Never-treated, excl. 2016	+1.895	0.724

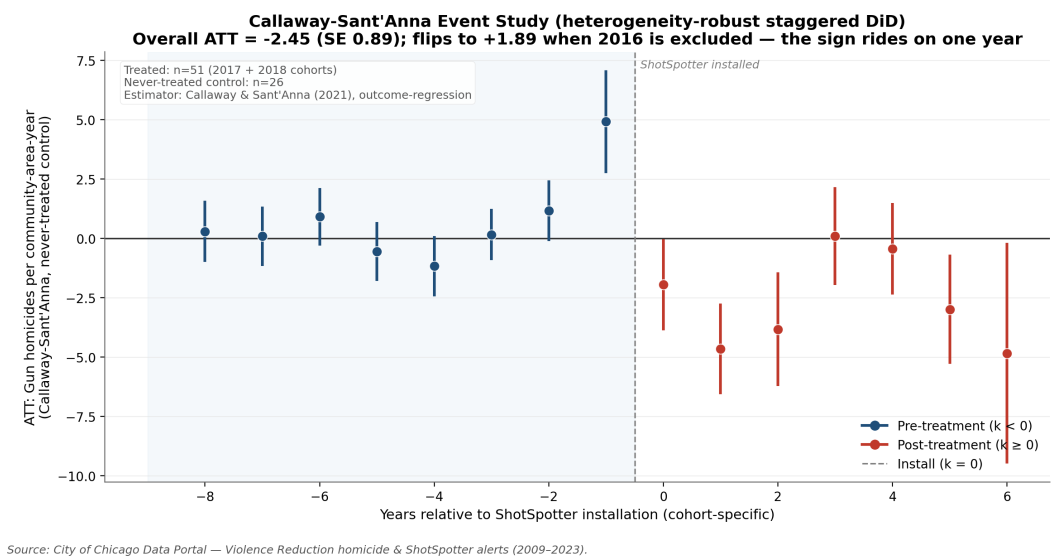

Figure 8: Callaway–Sant'Anna heterogeneity-robust event study. The negative overall ATT rides on the single 2016 pre-coefficient; dropping 2016 flips the overall estimate back to the sign of OLS.

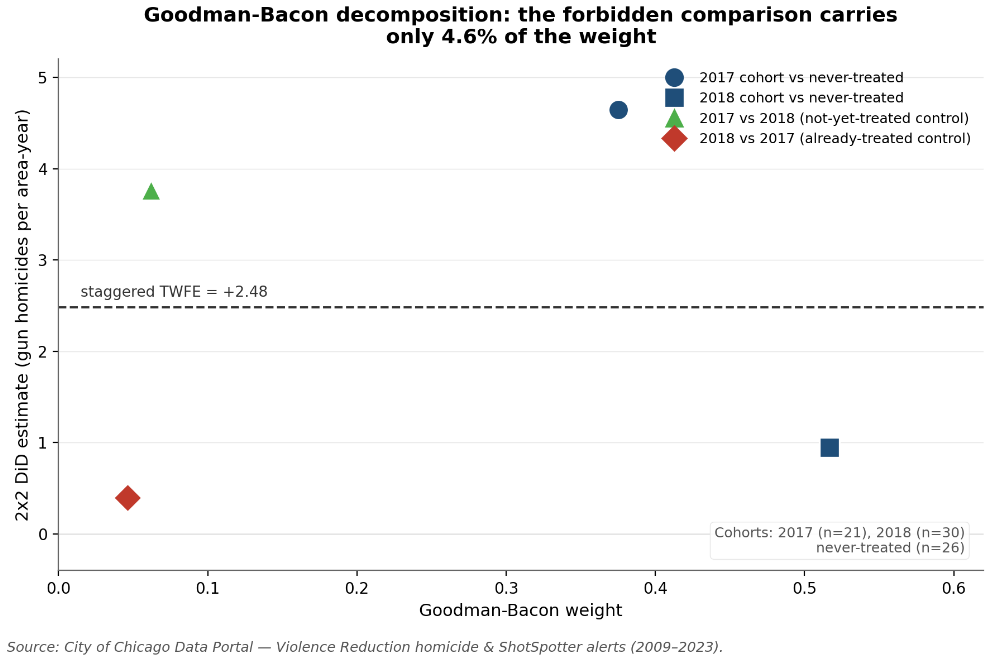

Figure 9: Goodman–Bacon decomposition of the staggered TWFE estimate. The "forbidden" already-treated comparison (red diamond) carries only 4.6% of the weight; the weighted 2×2 components reproduce the aggregate coefficient exactly. The estimate's instability is not staggered-timing contamination.

Table 6: Goodman–Bacon decomposition of the staggered two-way fixed-effects estimate (+2.481, SE 0.725). The weighted 2×2 components reproduce the aggregate exactly; the "forbidden" already-treated comparison carries only 4.6% of the weight.

Comparison	Weight	2×2 estimate
2017 cohort vs never-treated	0.375	+4.642
2018 cohort vs never-treated	0.517	+0.946
2017 vs 2018 (not-yet-treated control)	0.062	+3.765
2018 vs 2017 (already-treated control; forbidden)	0.046	+0.396

5.5 The heterogeneity-robust count estimator agrees

One loose end remains. The headline null is a Poisson (and Negative Binomial) two-way fixed-effects regression, which under a staggered rollout carries the same heterogeneity-weighting concern as linear TWFE—a two-way fixed-effects PPML can place negative weights on some treated cells

(Moreau-Kastler, 2025). And the heterogeneity-robust estimator used above, Callaway–Sant'Anna, is *linear*: it does not transfer to the multiplicative outcome the count argument itself calls for. The appropriate tool is a nonlinear, heterogeneity-robust count estimator, and we estimate it directly—a Poisson extended two-way fixed-effects model (Wooldridge, 2023) that saturates the treatment in cohort and calendar time, so each cohort-by-year effect is built from clean never-treated comparisons before aggregation, with a community-area cluster bootstrap for inference. It returns the same null: an overall ATT incidence-rate ratio of 1.05 (95% CI $[0.79, 1.33]$), with cohort-time effects hovering at one at every horizon (Figure 10, Table 7). The right tool removes the last escape hatch—the count-model null is not an artifact of the plain PPML's weighting.

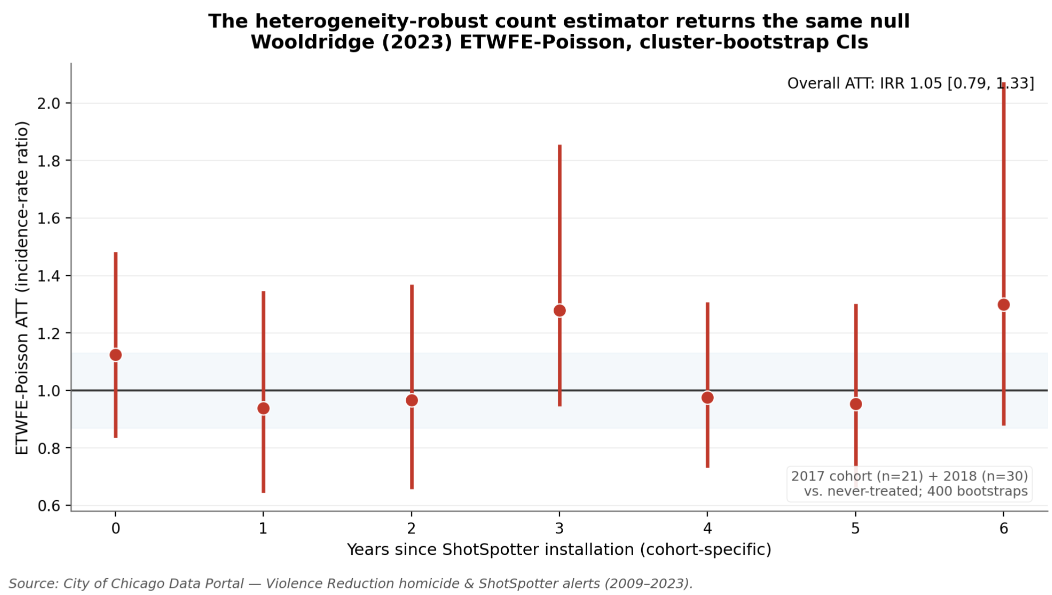

Figure 10: Wooldridge (2023) extended two-way fixed-effects Poisson event study—the heterogeneity-robust estimator for count outcomes under staggered adoption. Every cohort-time incidence-rate ratio hovers at one, and the overall ATT is 1.05 (95% CI $[0.79, 1.33]$, community-area cluster bootstrap). The count-model null is not a weighting artifact of the plain PPML.

Table 7: Wooldridge (2023) extended two-way fixed-effects Poisson: ATT incidence-rate ratios by years since install, and the cohort-size-weighted overall ATT. Community-area cluster bootstrap (400 draws). Every horizon and the overall estimate span one.

Years since install (k)	ATT (IRR)	95% CI
0	1.123	$[0.842, 1.475]$
1	0.938	$[0.650, 1.340]$
2	0.967	$[0.663, 1.362]$
3	1.279	$[0.951, 1.849]$
4	0.975	$[0.738, 1.301]$
5	0.953	$[0.668, 1.296]$
6	1.299	$[0.884, 2.067]$
Overall ATT	**1.048**	$[0.791, 1.330]$

5.6 The selection mechanism: what predicted placement

The paper has so far asserted that ShotSpotter was assigned because of pre-existing violence. Because that selection is the hinge of every identification problem here, we estimate it directly, modeling ShotSpotter placement across all 77 community areas as a function of pre-period (2009–2016) gun-homicide levels and the 2008–2012 Census socioeconomic indicators. One caveat governs the exercise: the city's siting decision was administrative and made at the police-district level—the OIG documents deployment across twelve districts—so community-area treatment is the neighborhood-level footprint of that coarser choice, and the model describes what that footprint tracks, not 77 independent siting decisions. Two facts emerge. First, placement tracks structural disadvantage even more tightly than the homicide count itself (Figure 11, Table 8): the hardship index separates treated from never-treated areas with an AUC of 0.93, and per-capita income and unemployment each with 0.92, against 0.79 for the pre-period gun-homicide count, and once hardship is held fixed the violence count adds essentially nothing (its coefficient is statistically insignificant). Violence and disadvantage are entangled—the same neighborhoods score high on both—so this does not say violence was irrelevant; it says the city sited ShotSpotter along the map of disinvestment, of which concentrated gun violence is one facet. The starkest summary of that map is not economic but biological: the ShotSpotter neighborhoods averaged 76 years of life expectancy in 2010 against 80 in the never-treated areas (Figure 11)—a four-year gap that states plainly who lives under the surveillance the city purchased.

Second, and decisively for identification, placement is nearly deterministic. A propensity model on pre-period violence, hardship, and income separates the two groups with an AUC of 0.94; the median propensity is 0.97 for ShotSpotter areas against 0.19 for never-treated areas; and the six highest-violence study neighborhoods all carry a placement propensity of at least 0.97—a band that *no* never-treated area reaches (Figure 12). The natural comparison—neighborhoods similar in pre-period violence that did not receive ShotSpotter—therefore does not exist in Chicago, which is exactly why the synthetic control below cannot be constructed. This is the selection-side statement of the counterfactual problem: the city installed ShotSpotter in essentially every highest-violence, highest-hardship neighborhood, leaving no comparably situated untreated unit from which to build a counterfactual for the units the study is about.

Placement also mapped onto the city's racial geography. ShotSpotter areas are on average 49% Black and 32% Hispanic, against 14% and 17% in never-treated areas, and the white, non-Hispanic share is the single sharpest separator in the whole analysis (13% versus 56%, AUC 0.95). The pattern survives conditioning: adding Black and Hispanic share to the placement model raises its AUC from 0.94 to 0.96, and in a linear-probability model with pre-period violence, hardship, and income, a one-SD rise in the Black share is associated with a +33 percentage-point change in placement probability ($p = 0.002$) and the Hispanic share +28 points ($p = 0.003$), while violence and hardship themselves turn insignificant (Table 9). Race, hardship, and violence are deeply entangled in a segregated city, and with 77 areas the design cannot separate them cleanly, so this is *not* evidence that race drove siting independently of disadvantage. What it shows, descriptively, is that ShotSpotter placement tracked Chicago's racial geography at least as tightly as it tracked violence or hardship. That coverage blankets the city's highest-Black-and-Latino police districts is already documented (MacArthur Justice Center, 2021; Office of Inspector General, 2021); our marginal contribution is the *conditional* result—that racial composition predicts placement even once pre-period violence and hardship are held fixed. (Community-area race is measured from the 2019–2023 ACS, which post-dates the rollout; because Chicago's neighborhood racial composition is highly persistent, we read it as a proxy for the pre-rollout map.)

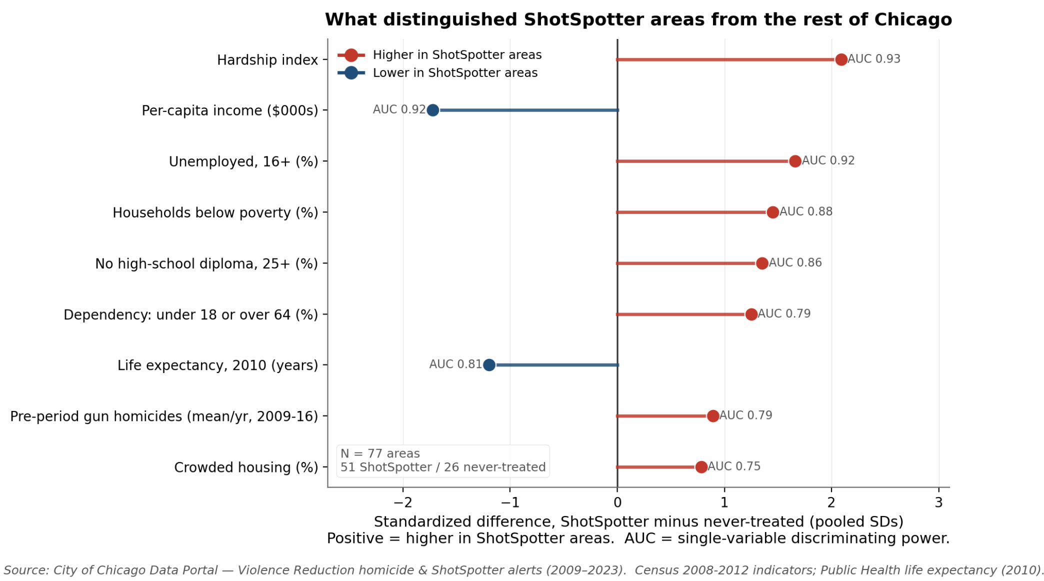

Figure 11: Standardized differences between ShotSpotter and never-treated community areas on each characteristic (treated minus control, in pooled SDs), with each variable's single-predictor discriminating power (AUC). Structural disadvantage—hardship, low income, unemployment—distinguishes the two groups more sharply than the pre-period gun-homicide count does.

Table 8: Balance: ShotSpotter (treated, $n = 51$) versus never-treated ($n = 26$) community areas. Std. diff. is the standardized mean difference (treated − control, in pooled SDs); AUC is the single-predictor discriminating power for placement. Structural disadvantage separates the groups more sharply than the pre-period gun-homicide count. Life expectancy is descriptive and is not entered in the placement regressions.

Characteristic	Treated	Control	Std. diff.	AUC
Hardship index	63.90	21.27	+2.09	0.931
Per-capita income ($000s)	18.65	39.11	−1.72	0.919
Unemployed, 16+ (%)	18.70	8.84	+1.66	0.919
Households below poverty (%)	26.44	12.60	+1.45	0.885
No high-school diploma, 25+ (%)	24.90	11.40	+1.35	0.860
Dependency: under 18 or over 64 (%)	38.42	30.51	+1.25	0.787
Life expectancy, 2010 (years)	76.16	80.43	−1.20	0.806
Pre-period gun homicides (mean/yr)	7.48	1.62	+0.89	0.793
Crowded housing (%)	5.84	3.12	+0.78	0.745
Racial composition (2019–2023 ACS)				
Black share (%)	48.9	13.5	+1.03	0.733
Hispanic/Latino share (%)	32.0	17.4	+0.55	0.537
White, non-Hispanic share (%)	12.6	55.7	−2.80	0.948

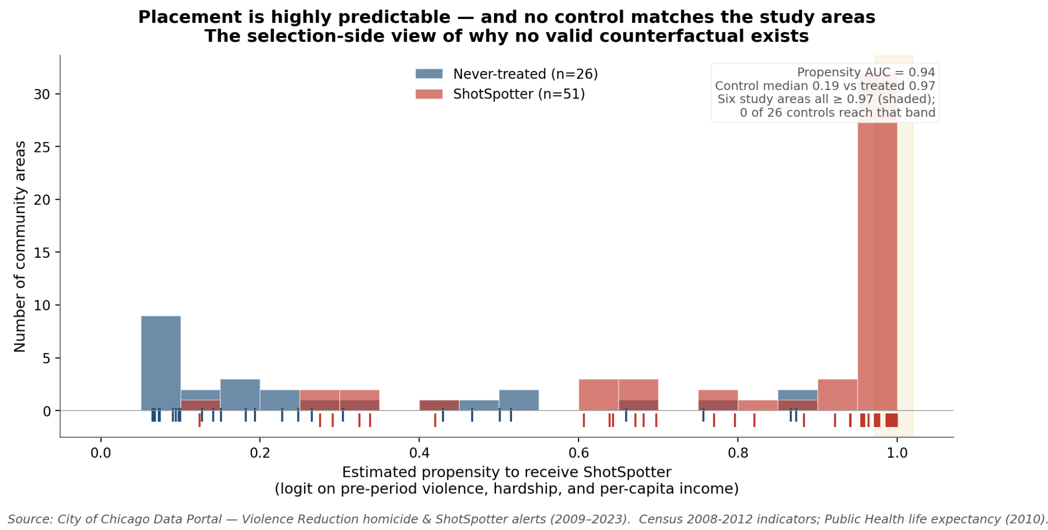

Figure 12: Estimated propensity to receive ShotSpotter, treated versus never-treated. The distributions are starkly separated (AUC 0.94); the shaded band holds the six highest-violence study neighborhoods (propensity ≥ 0.97), where no never-treated area appears. Placement is predictable enough that these units have no untreated counterpart—the selection-side view of why no valid counterfactual exists.

Table 9: Placement (linear-probability) models with standardized predictors and HC1 robust SEs (in parentheses); the dependent variable is receipt of ShotSpotter. Each coefficient is the change in placement probability per one-SD increase. The gun-homicide count does not predict placement once disadvantage is held fixed, while hardship and—conditionally—racial composition do.

Predictor	(1) Violence + hardship + income	(2) + racial composition
Pre-period gun homicides	$+0.016$	-0.023
	(0.026)	(0.027)
Hardship index	$+0.271^{***}$	$+0.123$
	(0.068)	(0.096)
Per-capita income	-0.065	$+0.012$
	(0.053)	(0.050)
Black share	—	$+0.333^{**}$
		(0.110)
Hispanic/Latino share	—	$+0.282^{**}$
		(0.094)
R^2 (LPM)	0.506	0.613
Logit AUC	0.935	0.956
N	77	77

** $p < 0.01$; *** $p < 0.001$. Predictors standardized to mean 0, SD 1.

5.7 Synthetic control is infeasible

For each of the six study neighborhoods, the synthetic-control optimizer concentrates 100% of the weight on a single donor—Washington Heights, the highest-violence non-ShotSpotter community area. Pre-treatment RMSEs are large (10–34 homicides per year against actual values of 15–70), meaning even the best linear combination of donors cannot reproduce the pre-treatment trajectory of any treated unit. The synthetic counterfactual lies far below the treated series in every panel of Figure 13, including in the pre-treatment years it was explicitly fit on.

Permutation inference makes the problem concrete rather than rescuing it. Running the same synthetic control on each of the 26 never-treated areas produces a tight placebo-gap distribution centered near zero against actual treated gaps of $+14$ to $+49$; the naive permutation p-value is below 0.001 for every treated unit. But this rejection is *invalid by construction*: it is dominated by the level mismatch between treated and donor units, not by any treatment effect. The 26 never-treated community areas occupy the low-violence end of Chicago's distribution, so there is no donor pool capable of constructing a credible counterfactual for the city's most violent neighborhoods. This is not a failure of method; it is a feature of the data. ShotSpotter's selection rule—deploy where violence is highest—is, in a literal sense, what makes synthetic control infeasible. The absence of a counterfactual is itself the finding.

The contrast with settings where synthetic control *does* work is instructive. In Kansas City, gunshot detection covers a localized 3.5-square-mile zone—about 1% of the city—leaving a large donor pool of untreated micro-units, and a microsynthetic control is feasible (Piza et al., 2024; Robbins & Davenport, 2021). Chicago is the mirror image: coverage blankets essentially every high-violence community area, so the pool of comparably violent untreated donors is empty. Infeasibility here is therefore not a shortcoming of the estimator but a consequence of near-universal treatment among the units of interest—the same fact the placement model states from the selection side. Permutation p-values from synthetic control lack a clean interpretation in any case (Ferman et al., 2020).

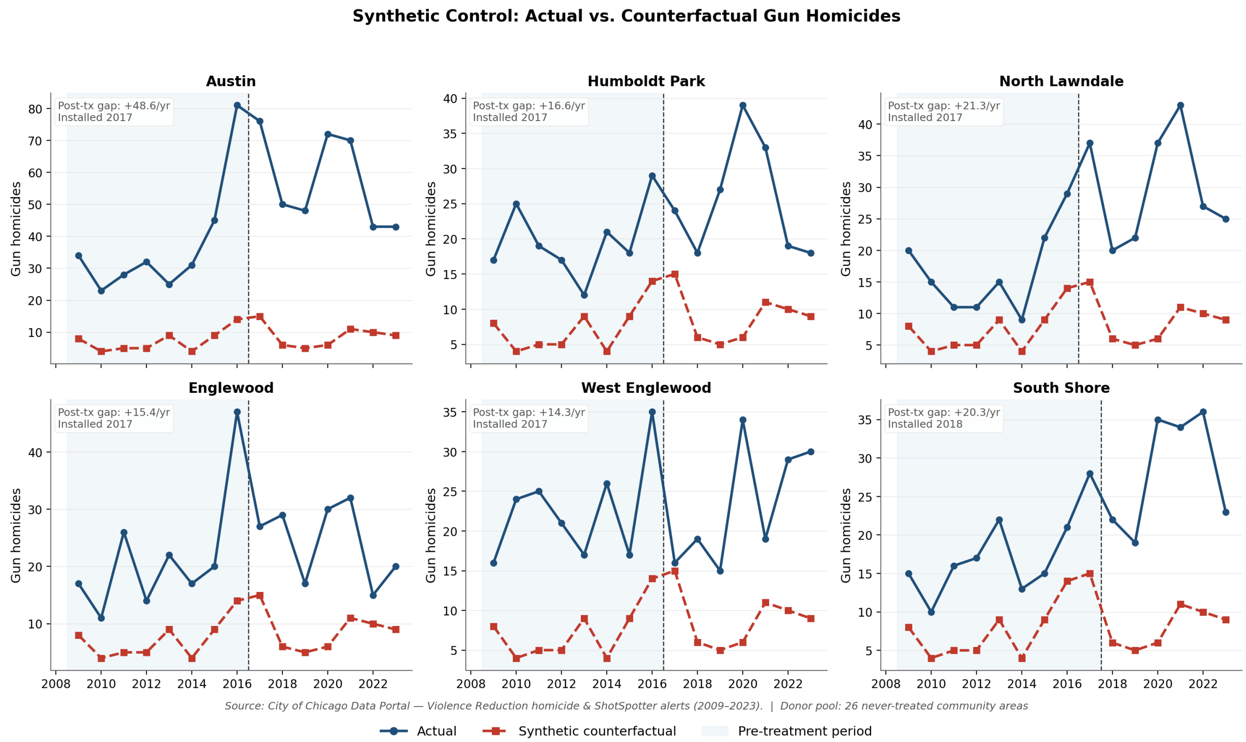

Figure 13: Synthetic control for the six study neighborhoods. The synthetic counterfactual (red) sits far below the actual series (blue) even in the pre-treatment window it was fit on, because no combination of the 26 lower-violence never-treated areas can match a high-violence treated unit. The method is infeasible for these data.

5.8 What does explain neighborhood-level violence?

The hardship index from the 2008–2012 Census ranks the six study neighborhoods at 55, 73, 85, 87, 89, and 94—against a city-wide hardship score below 50 and the lowest-violence community areas at 1–10. Per-capita income in these neighborhoods ranges from about $11,000 to $19,000, against a city-wide $28,202. The same neighborhoods that had the highest homicide rates *before* ShotSpotter still had the highest homicide rates after. Cross-sectional variation in violence tracks structural disadvantage tightly; cross-sectional variation in ShotSpotter coverage adds little explanatory power once disadvantage is held constant.

5.9 A second natural experiment: the September 2024 removal

Chicago switched ShotSpotter off on 22 September 2024, and the gun-homicide decline that followed has been read in public commentary as proof the technology never worked. The shutoff is a genuine second experiment, so it is worth asking whether it identifies anything the installation did not. It does not—and it fails in precisely the two ways the installation estimate fails, which is itself the cleanest confirmation of the paper's argument.

First, the common trend. Gun homicides had been falling city-wide from their 2021–2022 peak for two full years before the shutoff, in ShotSpotter and never-treated areas alike. On a common (indexed) scale the two groups decline in near-lockstep straight through the removal date (Figure 14): coverage-area homicides fell 37% from the pre- to the post-removal window, but never-treated areas fell 31% over the very same months. The naive "crime fell after ShotSpotter left" comparison attributes a city-wide trend to the shutoff, the exact inverse of the naive "crime rose after ShotSpotter arrived" comparison that would have credited the 2016–2021 rise to the rollout.

Second, the scale artifact. Netting out the common trend with a difference-in-differences does not rescue identification, because the additive DiD reproduces the very artifact of Section 5.2: it returns a "significant" removal *benefit* of -0.25 gun homicides per area-month ($p < 0.001$)—but only because coverage areas start from a far higher baseline (0.85 versus 0.20 gun homicides per area-month) and so move more in absolute terms for the same proportional change. On the correct multiplicative scale the removal effect is null (Poisson IRR 0.91, 95% CI $[0.54, 1.52]$; Table 10). The post window is also short—seven complete months, with 2025 only partially observed—so the estimate is under-powered on top of being non-identified. Both the raw drop and the additive DiD are mirages: these data can no more identify the removal effect than the installation effect, in either direction. The public debate has confidently drawn both unwarranted inferences.

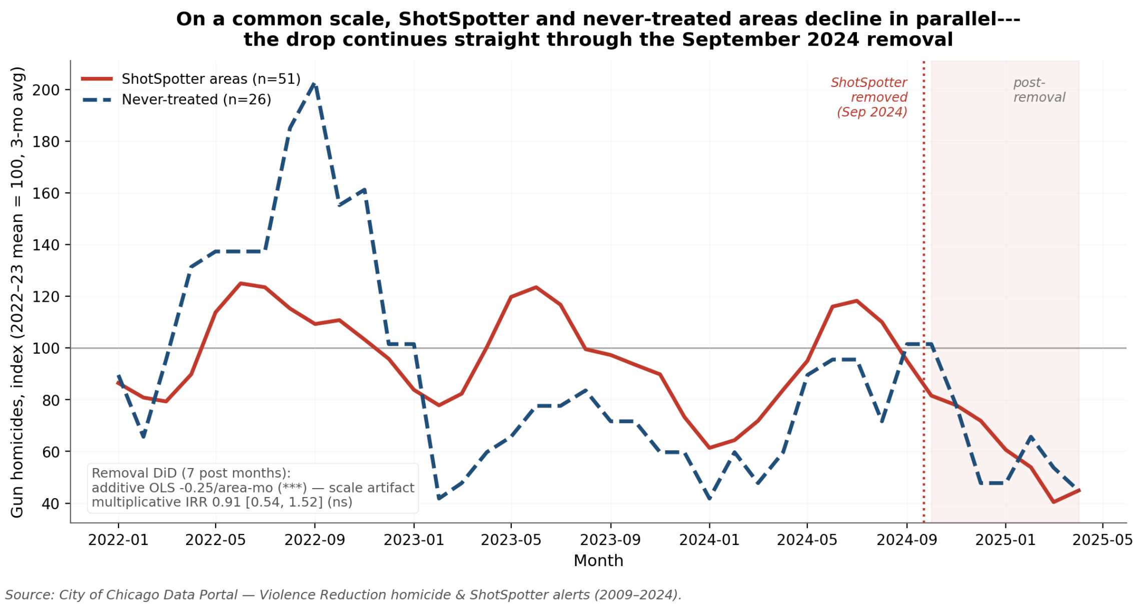

Figure 14: The September 2024 removal as a second natural experiment. Each series is indexed to its own 2022–2023 mean (= 100) and smoothed with a three-month moving average, so the two groups are compared on a common proportional scale. Gun homicides fall in near-lockstep in ShotSpotter and never-treated areas alike, straight through the removal date and into the post-removal window (shaded). The naive "crime fell after removal" claim mistakes a city-wide decline for a removal effect; the additive difference-in-differences that nets out that trend reproduces the paper's scale artifact (a "significant" −0.25/area-month that is null on the multiplicative scale, IRR 0.91).

Table 10: The September 2024 removal (monthly panel; seven complete post-removal months, 2025 partial). Both groups fall in near-identical proportion, so the naive drop is a city-wide trend; the additive DiD is "significant" only because coverage areas have a higher baseline, while the multiplicative model is null.

Panel A. Gun homicides per area-month, before vs. after the shutoff

Group	Pre	Post	% change
ShotSpotter areas ($n = 51$)	0.849	0.532	−37%
Never-treated ($n = 26$)	0.199	0.137	−31%

Panel B. Removal difference-in-differences (treated × post)

Estimator	Estimate (SE)	95% CI	p
OLS (additive, per area-month)	−0.255 (0.069)	—	< 0.001
Poisson (IRR)	0.909	$[0.545, 1.518]$	0.717

6 What the Data Can and Cannot Support

Taken together—(a) an OLS DiD coefficient that disappears under proper count-data regression (Poisson IRR 1.07, $p = 0.51$; Negative Binomial IRR 1.07, $p = 0.55$); (b) a Wald F-test that rejects parallel trends at the 1% level ($F = 19.9$, $p = 0.006$); (c) similarly significant pre-period placebos; (d) an infeasible synthetic control; and (e) a Callaway–Sant'Anna estimate whose sign reverses (-2.45 to $+1.90$) when one anomalous year is dropped—the evidence points to a single conclusion: **the design cannot credibly identify the causal effect of ShotSpotter installation on gun homicides.** And the September 2024 removal, examined the same way, reproduces both failures at once—a city-wide decline mistaken for a treatment effect, and an additive DiD that manufactures significance the multiplicative model erases (Section 5.9)—so the non-identification is a property of the data and the setting, not an artifact of how the installation event happens to be timed. Table 11 lays out why the estimators differ.

Table 11: Why the estimates differ—and what actually moves each one.

Estimator	Full panel	What actually moves it
Naive OLS DiD	$+2.63$ ($p < 0.001$)	Additive scale, inflated by high-count areas
Poisson / Neg. Binomial	IRR ≈ 1.07 (ns)	Positive in sign, but null on the multiplicative scale
Callaway–Sant'Anna	-2.45 ($p < 0.01$)	Flips to $+1.90$ once the 2016 outlier year is dropped

The data do not say ShotSpotter caused a 2.6-homicide-per-year *increase*—that statistic is a misspecification artifact. Neither do they say it caused a reduction—that reading takes a 2016-driven event-study coefficient too literally. The proper count-data specifications return statistically null effects with confidence intervals spanning the entire policy-relevant range.

We can, however, bound the answer in one direction: **the data are inconsistent with a large *deterrence* effect**—a broad, sustained fall in the number of shootings of the kind detection volume alone was meant to produce. A treatment that large, at these sample sizes, would have produced visibly negative post-treatment event-study coefficients with intervals excluding zero, robustness across specifications, and a cleaner signal in the higher-powered non-fatal-shootings analysis. None of these appear.

What the design cannot rule out is a benefit operating through a different channel. A boundary regression-discontinuity analysis attributes roughly 85 fewer deaths per year to faster emergency response inside coverage, working through victim survival rather than deterrence (University of Chicago Crime Lab, 2024; a working analysis, not yet peer-reviewed). The direct tests of that channel are equivocal rather than supportive: the two studies that measured case fatality found no reduction—Piza et al. (2024) compare the fatal and non-fatal shooting counts in the Kansas City coverage area and find no difference, and a Camden trauma-registry study found faster prehospital times but no significant adjusted mortality difference, albeit among more severely injured detected victims (Goldenberg et al., 2019). The mechanism is real—detection alerts precede the first 911 call by a median of roughly 90 seconds (Piza et al., 2023)—but its translation into fewer deaths is unproven. A mortality reduction of the magnitude the discontinuity implies would still fall *inside* our own count-model confidence interval, and a community-area-by-year homicide panel—which observes neither response times nor the conversion of shootings into deaths—is simply not built to detect it. Ruling out a large deterrence effect is therefore not the same as ruling out every benefit, and we do not claim the stronger statement. Whatever effect ShotSpotter had on gun homicides through deterrence was not large enough to dominate the noise, the 2016 anomaly, or the pre-existing trend differences across neighborhoods, and it falls far short of the reduction a $3 million detection contract was meant to deliver.

7 Why Detection Would Not Be Expected to Reduce Violence

Three explanations are consistent with this empirical pattern.

Selection. ShotSpotter was installed in the places that needed it most. Even if the technology produced a small benefit, the comparison would still look unfavorable because the rest of Chicago started from a much lower baseline with less room to deteriorate.

Detection is not intervention. The mechanism by which ShotSpotter could plausibly reduce homicides runs through faster police response and faster trauma triage. The data here do not measure response times, dispatch decisions, or trauma outcomes, and the 2021 OIG audit found that alerts rarely produced investigative evidence and that officer behavior did not consistently change in response to alerts (Office of Inspector General, 2021; Thompson, Lawrence, & La Vigne, 2020). If the operational link between detection and outcome is weak, detection volume is not a sufficient condition for fewer homicides.

Structural factors dominate. A surveillance technology cannot, by itself, change the distribution of poverty, unemployment, housing instability, or labor-market access. Programs that have reduced homicide in randomized or quasi-experimental evaluations—READI Chicago, Cure Violence, Becoming a Man—work on those structural channels directly. A microphone on a streetlight does not.

8 Limitations

The most important limitation is the one developed throughout the paper: the design itself cannot identify a clean treatment effect from these data, because the city did not generate the counterfactual we need—Chicago neighborhoods with ShotSpotter-eligible violence levels but no ShotSpotter installation. Beyond that, several caveats are worth naming explicitly. The unit of analysis is coarse. The crime-and-place literature favors micro-geographic units—street segments—over administrative geographies (Weisburd, 2015), and Schnell, Braga, and Piza (2017) show for Chicago specifically that community areas mask within-area heterogeneity that finer units capture; evaluations pinned to administrative units can also overstate coverage (Doucette et al., 2021). We use the community area deliberately, because it is the unit at which Chicago's open data are published and its policy debate is conducted, and the paper's claim is precisely that at this policy-grade resolution the effect is not identifiable—the coarseness is part of the finding, not an assumption to wave away. A micro-geographic re-estimation, once the city releases the sensor-coverage polygons, would be the natural complement. The panel is annual, masking within-year dynamics. Standard errors cluster on community area (about 77 clusters); a restricted wild-cluster bootstrap (Cameron, Gelbach, & Miller, 2008; 999 Rademacher replications) returns $p \approx 0.001$ for the OLS DiD in every specification (Appendix C, Table 14), so the headline coefficient is not an artifact of thin-cluster inference—the problem is the specification and the design, not the standard errors. Two event-study implementations are reported—a transparent hand-rolled difference-in-means kept for intuition, and the packaged Callaway–Sant'Anna estimator, which should be treated as the authoritative staggered-DiD result. The placement model is cross-sectional and descriptive: it documents the selection pattern but does not identify a causal siting rule, and its racial-composition measure is drawn from the 2019–2023 ACS, a proxy for the pre-rollout composition that relies on the persistence of Chicago's neighborhood racial geography. And the analysis observes neither police response times, dispatch decisions, nor arrests—the actual channel through which detection could plausibly reduce violence.

The comparison a credible design needs—similarly violent places without detection—must therefore be built outside the community-area frame, and the existing literature shows how: Connealy et al. (2024) construct it by matching Chicago police districts, and Doucette et al. (2021) by comparing metropolitan counties nationally; both recover the null this panel cannot identify. Within Chicago, three designs would do better with data the city has not released: a geographic regression discontinuity at sensor-coverage boundaries (which requires the unreleased coverage polygons), estimation at census-tract-by-month resolution, and direct measurement of the response-time mechanism around each activation date.

9 Conclusion

ShotSpotter in Chicago succeeded at detection: alerts climbed in covered areas because the technology measures gunfire that 911 calls underreport. What cannot be demonstrated is that detection translated into a reduction in lethal violence. Every design we ran on these data either produces a positive DiD coefficient that fails falsification and collapses under proper count-data modeling, an event study that violates parallel trends because of the 2016 anomaly, or a synthetic control whose donor pool cannot match treated-area violence levels. With no valid control group and a pre-period dominated by one outlier year, these data simply cannot identify a causal effect in either direction. They are inconsistent with a large *deterrence* effect, though not with the smaller, survival-channel mortality benefit a boundary discontinuity design attributes to faster response—an effect this panel is not built to see.

That non-result is itself a policy statement. The 2023 termination of Chicago's ShotSpotter contract is consistent with the absence, in the aggregate data, of the broad violence reduction the technology was marketed to deliver. Resources reallocated from detection contracts toward community-based violence-prevention programs—READI Chicago, Cure Violence, Becoming a Man—would, on the existing quasi-experimental evidence, have at least as good a chance of producing the homicide reductions ShotSpotter has not been shown to deliver at scale. Technology can help detect the problem; on this evidence, it has not been shown to solve it.

A The specification artifact recurs on enforcement outcomes

The count-data artifact of Section 5.2—a large, significant additive OLS coefficient that collapses to a null incidence-rate ratio under Poisson—is not specific to homicide counts. Running the *identical* two-way fixed-effects DiD on a completely different outcome reproduces it exactly, which is the cleanest available demonstration that the OLS coefficient measures a scale, not an effect.

Gunshot detection is documented to improve *process* outcomes—faster detection, more ballistic evidence, more gun recoveries—even where it does not reduce victimization (Piza et al., 2024). Chicago's process measures are FOIA-only, but the public analog is weapons-violation enforcement from the Crimes dataset (`ijzp-q8t2`). Running the same DiD on two enforcement measures—weapons-violation incidents and arrests, aggregated to the community-area-by-year panel—yields Figure 15, and it teaches two things.

First, the specification artifact is general. Weapons-violation incidents produce an OLS DiD of $+51.8$ per community-area-year ($p < 0.001$)—an order of magnitude larger than the homicide coefficient, and just as significant—yet it collapses to a statistically null incidence-rate ratio of 1.13 ($p = 0.48$) under Poisson with the same fixed effects; weapons arrests behave identically (OLS $+22.0$, $p < 0.01$; IRR 0.95, $p = 0.82$). The additive-scale inflation that produced the headline homicide coefficient is not a quirk of homicide counts—it recurs, unchanged, on a completely different outcome.

Second, on the substance, these data do not reproduce the process benefit that a microsynthetic control recovers in Kansas City (Piza et al., 2024): weapons enforcement shows no detectable relative increase under coverage once the outcome is modeled correctly. Whether that reflects the 2021 OIG finding that Chicago's alerts rarely produced investigative evidence (Office of Inspector General, 2021), or simply the same absent counterfactual that governs every estimate in this paper, these data cannot say. What can be said is double-edged: to whatever extent coverage concentrated enforcement, it did so in the lowest-life-expectancy, most heavily policed neighborhoods—the "procedural benefit" vendors advertise and the surveillance burden advocates warn of are two readings of the same coin.

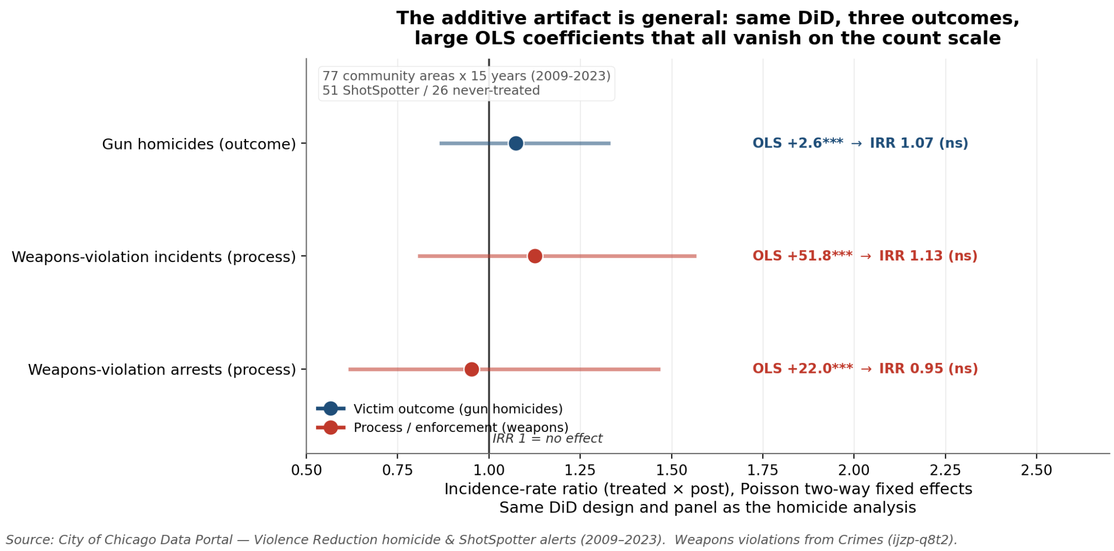

Figure 15: The same two-way fixed-effects DiD applied to a victim outcome (gun homicides) and two enforcement-process measures (weapons-violation incidents and arrests). Every outcome carries a large, significant OLS coefficient that collapses to a statistically null incidence-rate ratio on the count scale—the additive-scale artifact is not specific to homicides, and the process channel shows no detectable footprint of coverage in these community-area data.

Table 12: The same two-way fixed-effects DiD applied to a victim outcome and two enforcement-process outcomes, on the identical community-area-by-year panel. Every outcome carries a large, significant OLS coefficient that collapses to a null incidence-rate ratio on the count scale.

Outcome	OLS DiD	Poisson IRR	95% CI on IRR	p
Gun homicides (outcome)	$+2.627$	1.074	$[0.869, 1.328]$	0.508
Weapons-violation incidents (process)	$+51.78$	1.125	$[0.810, 1.563]$	0.481
Weapons-violation arrests (process)	$+21.96$	0.952	$[0.620, 1.464]$	0.824

B Cohort-specific event-study coefficients

Table 13 reports the aggregated cohort-specific event-study coefficients plotted in Figure 7. Every pre-treatment coefficient is large and negative, the pattern that the joint Wald test ($F = 19.9$, $p = 0.006$) formalizes as a rejection of parallel trends.

Table 13: Cohort-specific event study, aggregated ATT by event time k (years from install), sample-weighted, bootstrap CIs.

k (years from install)	ATT	95% CI
-7	-5.46	$[-9.30, -1.62]$
-6	-4.54	$[-8.53, -0.56]$
-5	-5.09	$[-8.97, -1.20]$
-4	-6.25	$[-10.04, -2.46]$
-3	-6.09	$[-10.01, -2.17]$
-2	-4.92	$[-8.92, -0.93]$
-1 (base)	0	—
0	-1.95	$[-6.57, +2.68]$
$+1$	-4.65	$[-8.79, -0.51]$
$+2$	-3.83	$[-8.12, +0.47]$
$+3$	$+0.10$	$[-4.85, +5.05]$
$+4$	-0.43	$[-5.25, +4.39]$
$+5$	-2.99	$[-7.19, +1.22]$

C Restricted wild-cluster bootstrap of the headline OLS DiD

Because the panel has only about 77 clusters, Table 14 re-tests the OLS DiD with a restricted wild-cluster bootstrap (Rademacher weights, null imposed, 999 replications; Cameron, Gelbach, & Miller, 2008). The coefficient stays significant in every specification, so the headline is not an artifact of thin-cluster inference—the problem is the additive specification and the design, not the standard errors.

Table 14: Restricted wild-cluster bootstrap of the headline OLS difference-in-differences. Cluster-robust t-statistics with a $G/(G-1)$ small-sample factor; the FWL-residualized implementation reproduces the statsmodels two-way-FE fit to 10^{-8}.

Specification	β (DiD)	t	Cluster-robust p	Wild-bootstrap p
Full panel	+2.627	4.29	< 0.001	0.001
Excl. COVID	+1.749	3.65	< 0.001	0.001
Excl. 2016	+3.189	4.35	< 0.001	0.001
Excl. COVID + 2016	+2.311	3.89	< 0.001	0.001

Data and Code Availability

Every input is a public product of the City of Chicago Data Portal (`https://data.cityofchicago.org`): Violence Reduction—Victims of Homicides and Non-Fatal Shootings (`gumc-mgzr`); Violence Reduction—ShotSpotter Alerts (Historical) (`3h7q-7mdb`); Census Data—Selected Socioeconomic Indicators, 2008–2012 (`kn9c-c2s2`); ACS 5-Year Data by Community Area (`t68z-cikk`); Public Health Statistics—Life Expectancy by Community Area (`qjr3-bm53`); Crimes—2001 to Present (`ijzp-q8t2`); and the Community Areas boundary file (`cauq-8yn6`). The complete analysis pipeline—data-processing, all five estimator classes, the placement model, and every figure—is openly available and reproduces the paper end to end at `https://github.com/abelma2/ACE592_Final_Project`.

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